Reduced Optimal Control with PDE Constraint: Model, Discretization, and First-Order Methods

Fachseminar

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Abstract

We study a linear-quadratic optimal control problem governed by the Poisson equation with homogeneous Dirichlet boundary conditions. The control acts as a volumetric source restricted to a tagged subdomain $\omega \subset \Omega$, and a Tikhonov term with parameter β regularizes the control in the $L^2(\Omega)$ metric. After P1 finite element discretization on a sequence of pre-generated meshes, we eliminate the state and work with the reduced functional $F(u)$, whose gradient is expressed through an adjoint solve and the control-space mass matrix.

On each mesh we compute the minimizer u_* in two ways: (i) by solving the symmetric positive definite stationarity system with the Conjugate Gradient method (CG), and (ii) by applying first-order optimization methods to the reduced problem, namely Barzilai–Borwein (BB), fixed-step gradient descent with step size $1/L$, and a constant-parameter Nesterov scheme based on (m, L). We report convergence histories in terms of the U-norm of the reduced gradient, as well as a mesh-independence study comparing objective values, norms of u_* and $y(u_*)$, and iteration counts across mesh sizes.

The results show stable optimal values as the mesh is refined and highlight the efficiency advantage of CG for the quadratic reduced problem, while BB and accelerated gradient methods provide competitive first-order baselines.

1 Problem Statement

We consider a linear-quadratic optimal control problem governed by the Poisson equation on a polygonal domain $\Omega \subset \mathbb{R}^2$ (in computations, typically $[0,1]^2$). A circular subregion $\omega \subset \Omega$ with nonzero Lebesgue measure is tagged on the mesh and serves as the support of the control. The control acts as a volumetric source term restricted to ω:

$$-\Delta y = \chi_\omega \, u \quad \text{in } \Omega, \tag{1}$$

$$y = 0 \quad \text{on } \partial\Omega, \tag{2}$$

where χ_ω denotes the characteristic function of ω. We denote by $V = H_0^1(\Omega)$ the state space and by $U = L^2(\Omega)$ the control space. For a given target $y_d \in L^2(\Omega)$ and a regularization parameter $\beta > 0$, we minimize

$$J(y,u) = \tfrac{1}{2}\int_\Omega (y - y_d)^2 \, dx + \tfrac{\beta}{2}\int_\Omega u^2 \, dx. \tag{3}$$

The target is chosen as the pyramid-like profile

$$y_d(x) = 0.5 - \max\left(|x_1 - 0.5|,\, |x_2 - 0.5|\right), \tag{4}$$

which belongs to $L^2(\Omega)$. The control enters only on ω via χ_ω, so the effective source equals u on ω and vanishes on $\Omega \setminus \omega$. The PDE with homogeneous Dirichlet boundary conditions is well-posed for $u \in L^2(\Omega)$, yielding a unique weak solution $y \in V$ by Lax–Milgram.

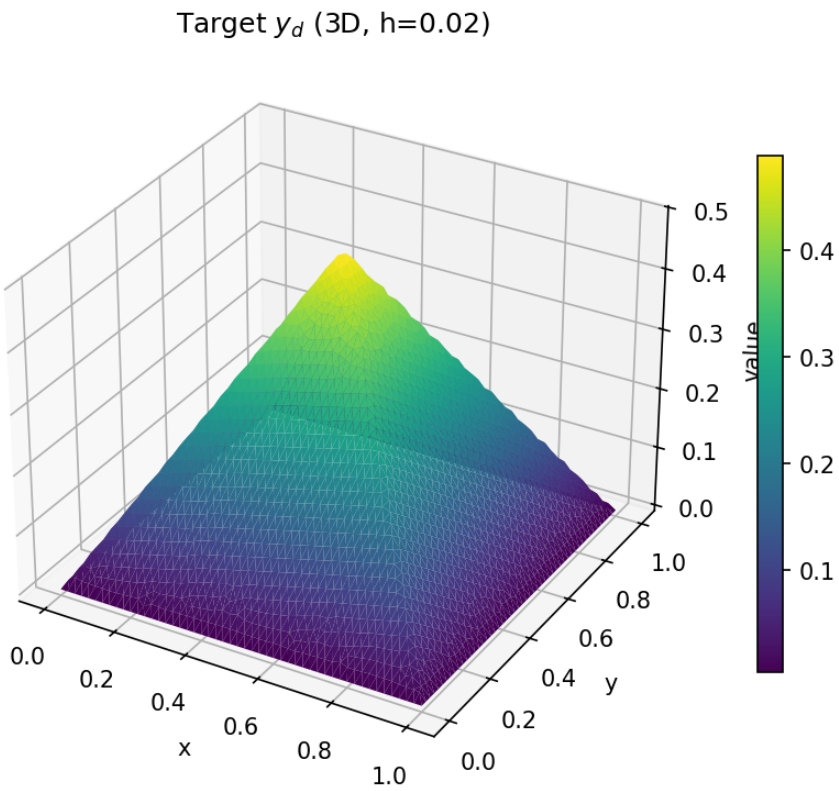

Figure 1: 3D surface of the target profile y_d on Ω.

2 Variational Formulation and Discretization

Weak formulation. Define the bilinear and linear forms

$$a(y,v) = \int_\Omega \nabla y \cdot \nabla v \, dx \quad \text{for } y, v \in V, \qquad b(u,v) = \int_\omega u \, v \, dx \quad \text{for } v \in U, v \in V. \tag{5}$$

For any $u \in L^2(\Omega)$, the weak state equation reads: find $y \in V$ such that

$$a(y,v) = b(u,v) \quad \forall v \in V. \tag{6}$$

By continuity and coercivity of $a(\cdot,\cdot)$ on V and boundedness of $b(\cdot,\cdot)$, this admits a unique $y \in V$.

Finite elements. Let $\mathcal{T}_h$ be a conforming triangulation of Ω. We use the conforming P1 space $V_h \subset V$ with nodal basis $\{\varphi_i\}_{i=1}^n$. The control is represented in the same P1 basis but measured with the L^2 inner product ("control metric"). We assemble the matrices

$$A_{ij} = \int_\Omega \nabla\varphi_j \cdot \nabla\varphi_i \, dx, \quad M_{ij} = \int_\Omega \varphi_j \, \varphi_i \, dx, \quad B_{ij} = \int_\omega \varphi_j \, \varphi_i \, dx, \quad (M_U)_{ij} = \int_\Omega \varphi_j \, \varphi_i \, dx. \tag{7}$$

Here, A is the stiffness matrix for state/adjoint solves, M is the state mass matrix used to weight the tracking term, B maps control coefficients to the state RHS via restriction to ω, and M_U is the control-space mass matrix defining the L^2 inner product in U_h.

Boundary conditions. Homogeneous Dirichlet boundary conditions are imposed on the state/adjoint by modifying A and setting the RHS at boundary degrees of freedom to zero when solving. Importantly, no boundary conditions are applied to B or M_U; these operators remain the unmodified L^2-based forms on Ω.

Discrete state equation and objective. With coefficient vectors $\mathbf{y}, \mathbf{u} \in \mathbb{R}^n$ and target coefficients $\mathbf{y}_d \in \mathbb{R}^n$, the discrete state equation and cost read

$$A\mathbf{y} = B\mathbf{u}, \tag{8}$$

$$J(\mathbf{y}, \mathbf{u}) = \tfrac{1}{2}(\mathbf{y} - \mathbf{y}_d)^\top M(\mathbf{y} - \mathbf{y}_d) + \tfrac{\beta}{2}\mathbf{u}^\top M_U \mathbf{u}. \tag{9}$$

3 Reduced Formulation

State elimination and reduced functional. The discrete PDE constraint $A\mathbf{y} = B\mathbf{u}$ defines the linear control-to-state map $\mathbf{y}(\mathbf{u}) = A^{-1}B\mathbf{u}$. Substituting into the discrete objective yields the reduced functional

$$F(\mathbf{u}) = \tfrac{1}{2}(\mathbf{y}(\mathbf{u}) - \mathbf{y}_d)^\top M(\mathbf{y}(\mathbf{u}) - \mathbf{y}_d) + \tfrac{\beta}{2}\mathbf{u}^\top M_U \mathbf{u}. \tag{10}$$

Derivative vs. U-gradient (Riesz representative). We equip the discrete control space $U_h \cong \mathbb{R}^n$ with the inner product

$$\langle \mathbf{a}, \mathbf{b} \rangle_U := \mathbf{a}^\top M_U \mathbf{b}, \qquad \|\mathbf{a}\|_U := \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle_U}. \tag{11}$$

The (Fréchet) derivative $F'(\mathbf{u})$ is a linear functional on U_h, i.e., $F'(\mathbf{u}) \in U_h^*$. The U-gradient $\nabla_U F(\mathbf{u}) \in U_h$ is defined as the Riesz representative of $F'(\mathbf{u})$ with respect to $\langle \cdot, \cdot \rangle_U$:

$$\langle \nabla_U F(\mathbf{u}), \mathbf{v} \rangle_U = F'(\mathbf{u})[\mathbf{v}] \qquad \forall \mathbf{v} \in \mathbb{R}^n. \tag{12}$$

Connection to ∇F in coefficient coordinates. Sometimes it is convenient to identify the derivative $F'(\mathbf{u})$ with a vector via the standard Euclidean inner product $\langle \mathbf{a}, \mathbf{b} \rangle_2 = \mathbf{a}^\top \mathbf{b}$. We denote this coordinate gradient by $\nabla F(\mathbf{u}) \in \mathbb{R}^n$, defined through

$$F'(\mathbf{u})[\mathbf{v}] = \langle \nabla F(\mathbf{u}), \mathbf{v} \rangle_2 = \nabla F(\mathbf{u})^\top \mathbf{v} \qquad \forall \mathbf{v} \in \mathbb{R}^n. \tag{13}$$

Comparing with the defining identity for $\nabla_U F$ gives the Riesz-map relation

$$\nabla F(\mathbf{u}) = M_U \nabla_U F(\mathbf{u}). \tag{14}$$

Adjoint formula and explicit expression. Using the adjoint variable $\mathbf{p}$ defined by

$$A\mathbf{p} = M(\mathbf{y}(\mathbf{u}) - \mathbf{y}_d), \tag{15}$$

one obtains for all directions $\mathbf{v}$

$$F'(\mathbf{u})[\mathbf{v}] = (B^\top \mathbf{p} + \beta M_U \mathbf{u})^\top \mathbf{v}. \tag{16}$$

Therefore, the U-gradient (i.e., the Riesz representative in the M_U-metric) is

$$\nabla_U F(\mathbf{u}) = M_U^{-1} B^\top \mathbf{p} + \beta \mathbf{u}. \tag{17}$$

The optimality condition is

$$\nabla_U F(\mathbf{u}_*) = \mathbf{0} \quad \Longleftrightarrow \quad B^\top \mathbf{p}_* + \beta M_U \mathbf{u}_* = \mathbf{0}. \tag{18}$$

Quadratic structure and stationarity system. Since $\mathbf{y}(\mathbf{u})$ depends linearly on $\mathbf{u}$, the reduced problem is a strictly convex quadratic program. Indeed, with the control-to-state operator $S := A^{-1}B$ we have

$$F(\mathbf{u}) = \tfrac{1}{2}(S\mathbf{u} - \mathbf{y}_d)^\top M(S\mathbf{u} - \mathbf{y}_d) + \tfrac{\beta}{2}\mathbf{u}^\top M_U \mathbf{u} \tag{19}$$

$$= \tfrac{1}{2}\mathbf{u}^\top (S^\top M S + \beta M_U)\mathbf{u} - \left(S^\top M \mathbf{y}_d\right)^\top \mathbf{u} + \tfrac{1}{2}\mathbf{y}_d^\top M \mathbf{y}_d. \tag{20}$$

Using $S = A^{-1}B$ and symmetry of A, this can be written as the unconstrained quadratic program

$$\min_{\mathbf{u}\in\mathbb{R}^n} \tfrac{1}{2}\mathbf{u}^\top Q\mathbf{u} - \mathbf{b}^\top \mathbf{u} + \text{const}, \qquad Q := B^\top A^{-1} M A^{-1} B + \beta M_U, \quad \mathbf{b} := B^\top A^{-1} M \mathbf{y}_d. \tag{21}$$

Since $B^\top A^{-1} M A^{-1} B$ is symmetric positive semidefinite and βM_U is symmetric positive definite for $\beta > 0$, it follows that Q is symmetric positive definite, hence F is strictly convex. Here the constant term is const $= \frac{1}{2}\mathbf{y}_d^\top M \mathbf{y}_d$, which does not depend on $\mathbf{u}$ and therefore does not influence the minimizer. Eliminating $\mathbf{y}$ and $\mathbf{p}$ from the optimality condition yields the symmetric positive definite (SPD) linear system

$$Q\mathbf{u}_* = \mathbf{b}. \tag{22}$$

Lipschitz constant L and strong convexity modulus m. The constants L and m quantify smoothness and strong convexity of the reduced functional in the control metric. More precisely, L is the Lipschitz constant of the U-gradient,

$$\|\nabla_U F(\mathbf{u}) - \nabla_U F(\mathbf{v})\|_U \le L\|\mathbf{u} - \mathbf{v}\|_U \qquad \forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^n, \tag{23}$$

and $m > 0$ is the strong convexity modulus,

$$F(\mathbf{v}) \ge F(\mathbf{u}) + \langle \nabla_U F(\mathbf{u}), \mathbf{v} - \mathbf{u}\rangle_U + \tfrac{m}{2}\|\mathbf{v} - \mathbf{u}\|_U^2 \qquad \forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^n. \tag{24}$$

For the quadratic reduced problem, the U-gradient's U-Hessian is

$$H\mathbf{v} = \nabla_U^2 F(\mathbf{u})\mathbf{v} = M_U^{-1} Q\mathbf{v} = M_U^{-1} B^\top A^{-1} M A^{-1} B\mathbf{v} + \beta\mathbf{v}. \tag{25}$$

This operator is self-adjoint and SPD with respect to $\langle\cdot,\cdot\rangle_U$. Accordingly, m and L are the extreme generalized eigenvalues of the pencil (Q, M_U),

$$Q\mathbf{v} = \lambda M_U \mathbf{v}, \qquad m = \lambda_{\min},\ L = \lambda_{\max}, \tag{26}$$

equivalently the extreme eigenvalues of the SPD matrix $M_U^{-1/2} Q M_U^{-1/2}$. Their ratio $\kappa = L/m$ is the condition number that governs convergence rates of GD and Nesterov in the next section.

4 Optimization Methods

Unless stated otherwise, we initialize at $\mathbf{u}_0 = \nabla_U F(\mathbf{0})$ (as in the code) and stop when $\|\nabla_U F(\mathbf{u}_k)\|_U \le \texttt{tol}_{abs} + \texttt{tol}_{rel}\|\nabla_U F(\mathbf{0})\|_U$ with $\texttt{tol}_{abs} = 10^{-5}$ and $\texttt{tol}_{rel} = 10^{-3}$ or a maximum number of 1000 iterations is reached.

4.1 Steepest Descent/ Gradient method with constant Stepsize $\alpha = 1/L$ (GD)

For an L-Lipschitz gradient in the U-metric, the fixed-step update

$$\mathbf{u}_{k+1} = \mathbf{u}_k - \alpha\,\nabla_U F(\mathbf{u}_k), \quad \alpha = 1/L. \tag{27}$$

Algorithm 1: Steepest Descent with constant stepsize $\alpha = 1/L$

Data: tolerance `tol` > 0, `max_iter`, model with $\nabla_U F$, $\|\cdot\|_U$
Result: sequence $(\mathbf{u}_k)$ converging to a minimizer
$\mathbf{u} \leftarrow \mathbf{u}_0 = \nabla_U F(\mathbf{0})$
Estimate L by power iteration; $\alpha \leftarrow 1/L$;
for $k = 0, 1, \ldots$ **do**
  $\mathbf{g} \leftarrow \nabla_U F(\mathbf{u})$; $g_n \leftarrow \|\mathbf{g}\|_U$;
  if $g_n \leq$ *tol* **then**
    break
  end
  $\mathbf{u} \leftarrow \mathbf{u} - \alpha\, \mathbf{g}$;
end

4.2 Steepest Descent with Barzilai-Borwein stepsize strategy (BB)

BB methods approximate the inverse curvature along the last step by selecting the steplength from secant information in the U-inner product. Define $\mathbf{s}_k = \mathbf{u}_k - \mathbf{u}_{k-1}$, $\mathbf{d}_k = \nabla_U F(\mathbf{u}_k) - \nabla_U F(\mathbf{u}_{k-1})$, and the U-inner product $\langle \cdot, \cdot \rangle_U$. The two classical choices are

$$\alpha_k^{\mathrm{BB1}} = \frac{\langle \mathbf{d}_k, \mathbf{d}_k \rangle_U}{\langle \mathbf{s}_k, \mathbf{d}_k \rangle_U}, \qquad \alpha_k^{\mathrm{BB2}} = \frac{\langle \mathbf{s}_k, \mathbf{s}_k \rangle_U}{\langle \mathbf{s}_k, \mathbf{d}_k \rangle_U}. \tag{28}$$

We alternate BB1/BB2 per iteration and fall back to $\alpha_k = 1$ if $\left|\langle \mathbf{s}_k, \mathbf{d}_k \rangle_U\right|$ is numerically tiny. The update is $\mathbf{u}_{k+1} = \mathbf{u}_k - (1/\alpha_k)\, \nabla_U F(\mathbf{u}_k)$.

Algorithm 2: Steepest Descent with Barzilai-Borwein stepsize strategy

Data: `tol` > 0, `max_iter`, model with $\nabla_U F$, $\langle \cdot, \cdot \rangle_U$, $\|\cdot\|_U$
Result: sequence $(\mathbf{u}_k)$
$\mathbf{u}_{-1} \leftarrow \mathbf{0}$; $\mathbf{g}_{-1} \leftarrow \nabla_U F(\mathbf{0})$;
$\mathbf{u}_0 \leftarrow \mathbf{u}_0 = \nabla_U F(\mathbf{0})$;
$\mathbf{g}_0 \leftarrow \nabla_U F(\mathbf{u}_0)$
for $k = 0, 1, \ldots$ **do**
  if $\|\mathbf{g}_k\|_U \leq$ *tol* **then**
    break
  end
  $\mathbf{s}_k \leftarrow \mathbf{u}_k - \mathbf{u}_{k-1}$; $\mathbf{d}_k \leftarrow \mathbf{g}_k - \mathbf{g}_{k-1}$; $sd \leftarrow \langle \mathbf{s}_k, \mathbf{d}_k \rangle_U$;
  if $|sd| < \varepsilon$ **then**
    $\alpha_k \leftarrow 1$;
  else
    if *k is even* **then**
      $\alpha_k \leftarrow \langle \mathbf{d}_k, \mathbf{d}_k \rangle_U / sd$;
    else
      $\alpha_k \leftarrow sd / \max\{\langle \mathbf{s}_k, \mathbf{s}_k \rangle_U, \varepsilon\}$;
    end
  end
  $\mathbf{u}_{k+1} \leftarrow \mathbf{u}_k - (1/\alpha_k)\, \mathbf{g}_k$;
  $\mathbf{u}_{k-1} \leftarrow \mathbf{u}_k$; $\mathbf{g}_{k-1} \leftarrow \mathbf{g}_k$;
  $\mathbf{u}_k \leftarrow \mathbf{u}_{k+1}$; $\mathbf{g}_k \leftarrow \nabla_U F(\mathbf{u}_k)$;
end

4.3 Nesterov Acceleration with constant parameters

Set

$$\alpha = 1/L, \qquad \kappa = L/m, \qquad \beta = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}.$$

Then the constant-parameter Nesterov scheme reads

$$\mathbf{y}_k = \mathbf{u}_k + \beta(\mathbf{u}_k - \mathbf{u}_{k-1}), \qquad \mathbf{u}_{k+1} = \mathbf{y}_k - \alpha \nabla_U F(\mathbf{y}_k). \tag{29}$$

Algorithm 3: Nesterov Acceleration with constant parameters

Data: `tol` > 0, `max_iter`, model with $\nabla_U F$, M_U, Hessian action H
Result: sequence $(\mathbf{u}_k)$
Estimate (L, m) from generalized eigenpairs of (Q, M_U); $\alpha \leftarrow 1/L$; $\kappa \leftarrow L/m$;
$\beta \leftarrow (\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)$
$\mathbf{u}_{-1} \leftarrow \mathbf{u}_0$; $\mathbf{u} \leftarrow \mathbf{u}_0$;
for $k = 0, 1, \ldots$ **do**
  $\mathbf{y} \leftarrow \mathbf{u} + \beta(\mathbf{u} - \mathbf{u}_{k-1})$;
  $\mathbf{g} \leftarrow \nabla_U F(\mathbf{y})$; $g_n \leftarrow \|\mathbf{g}\|_U$;
  if $g_n \le$ *tol* **then**
    break
  end
  $\mathbf{u}_{k+1} \leftarrow \mathbf{y} - \alpha\, \mathbf{g}$;
end

4.4 Conjugate Gradient method (CG) for the Stationarity System

The reduced problem is strictly convex and quadratic. The first-order optimality condition $\nabla_U F(\mathbf{u}_*) = 0$ can be written as an SPD linear system in the control unknowns,

$$Q\mathbf{u}_* = B^\top A^{-1} M \mathbf{y}_d, \qquad Q := B^\top A^{-1} M A^{-1} B + \beta M_U. \tag{30}$$

In the implementation, we solve this system with the Conjugate Gradient method in a matrix-free manner (Hessian action only). Concretely, we apply $Q\mathbf{v}$ via the reduced Hessian action and the control metric, and run the standard CG recurrences while tracking the residual. We use a standard relative residual stopping criterion (as in SciPy's `scipy.sparse.linalg.cg`): with residual $\mathbf{r}_k := \mathbf{b} - Q\mathbf{u}_k$, we stop once

$$\|\mathbf{r}_k\|_2 \le \texttt{rtol}\, \|\mathbf{b}\|_2,$$

and we set $\texttt{rtol} = 10^{-5}$ in all runs. Moreover, since $\nabla_U F(\mathbf{u}) = M_U^{-1}(Q\mathbf{u} - \mathbf{b}) = -M_U^{-1}\mathbf{r}$, the reduced gradient norm can be obtained directly from the residual via

$$\|\nabla_U F(\mathbf{u}_k)\|_U = \sqrt{\mathbf{r}_k^\top M_U^{-1} \mathbf{r}_k},$$

which we record as the convergence curve in Figure 2.

4.5 Remarks on Metrics and Implementation

All inner products and norms above are in the control metric induced by M_U. This choice ensures that steplengths and convergence tests reflect the natural topology of the control space (consistent with regularization). The adjoint reuse of A facilitates efficient linear solves and preconditioning.

Algorithm 4: CG for $Q\mathbf{u} = \mathbf{b}$ (matrix-free)

Data: model with A^{-1} solves, matrices B, M, M_U, Hessian action H; $\texttt{rtol} > 0$
Result: $\mathbf{u}_*$
$\mathbf{b} \leftarrow B^\top A^{-1}(M\mathbf{y}_d)$;
Define operator $Q\mathbf{v} := M_U(H\mathbf{v})$ (equivalently $B^\top A^{-1} M A^{-1} B\mathbf{v} + \beta M_U \mathbf{v}$);
Call CG to solve $Q\mathbf{u} = \mathbf{b}$ to relative tolerance $\texttt{rtol} = 10^{-5}$;

5 Numerical Results

We report results on a representative external mesh with size $h = 0.02$ (here $n = 3093$ control degrees of freedom). We use control region tag $\omega = 2$ and Tikhonov parameter $\beta = 10^{-3}$. The first-order methods (BB, GD, Nesterov) are initialized at $\mathbf{u}_0 = \nabla_U F(\mathbf{0})$ and stopped when $\|\nabla_U F(\mathbf{u}_k)\|_U \leq \texttt{tol}_{abs} + \texttt{tol}_{rel}\|\nabla_U F(\mathbf{0})\|_U$ with $\texttt{tol}_{abs} = 10^{-5}$ and $\texttt{tol}_{rel} = 10^{-3}$.

For GD, the fixed step uses an estimated Lipschitz constant L (power iteration on the Hessian action); on this mesh we obtain $L \approx 1.00028 \times 10^{-3}$. As an additional method tailored to the quadratic structure, we also run CG on the SPD stationarity system $Q\mathbf{u} = \mathbf{b}$.

Method	Final cost $F(\mathbf{u}_k)$	Iterations
CG ($Q\mathbf{u} = \mathbf{b}$)	0.0161709	6
BB	0.0161709	2
GD ($1/L$)	0.0161709	6
Nesterov	0.0161708	3

Table 1: Representative-mesh results ($h = 0.02$). All methods reach the same optimum up to numerical precision; BB attains the stopping criterion in the fewest first-order iterations, while CG solves the stationarity system in a modest number of Krylov steps.

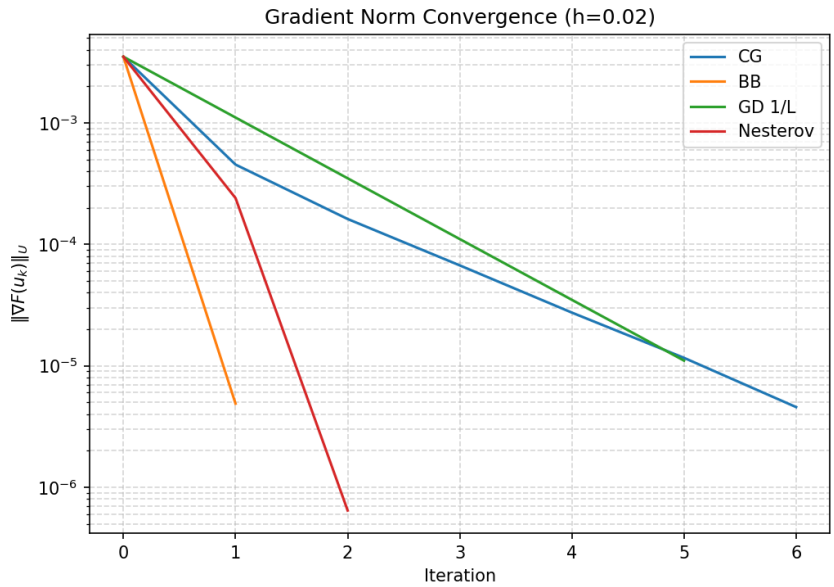

Figure 2: Convergence of gradient norm for a representative mesh ($h = 0.02$). The control-space norm $\|\cdot\|_U$ is induced by M_U.

Overall, Figure 2 shows comparable decay of $\|\nabla_U F(u_k)\|_U$ across the methods, and Table 1 confirms agreement in the final objective value. The optimized fields (Figure 3) are shown for the CG solution.

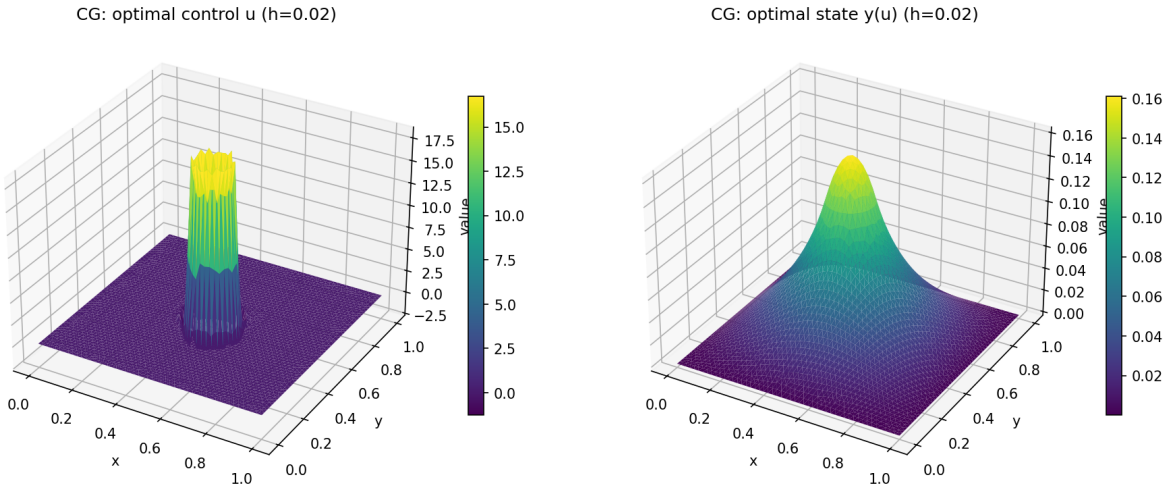

Figure 3: Optimized control u_* (left) and the corresponding state $y(u_*)$ (right) computed by CG on the stationarity system, shown for a representative mesh ($h = 0.02$).

6 Mesh Independence

We study how the discretized solution and the observed optimizer performance depend on mesh resolution. We use a fixed family of *pre-generated* triangular meshes of $\Omega = [0, 1]^2$ with a circular control region ω (physical tag $\omega = 2$) for $h \in \{0.06, 0.04, 0.03, 0.02, 0.015, 0.012, 0.01, 0.008\}$; the meshes are visualized in Figure 4.

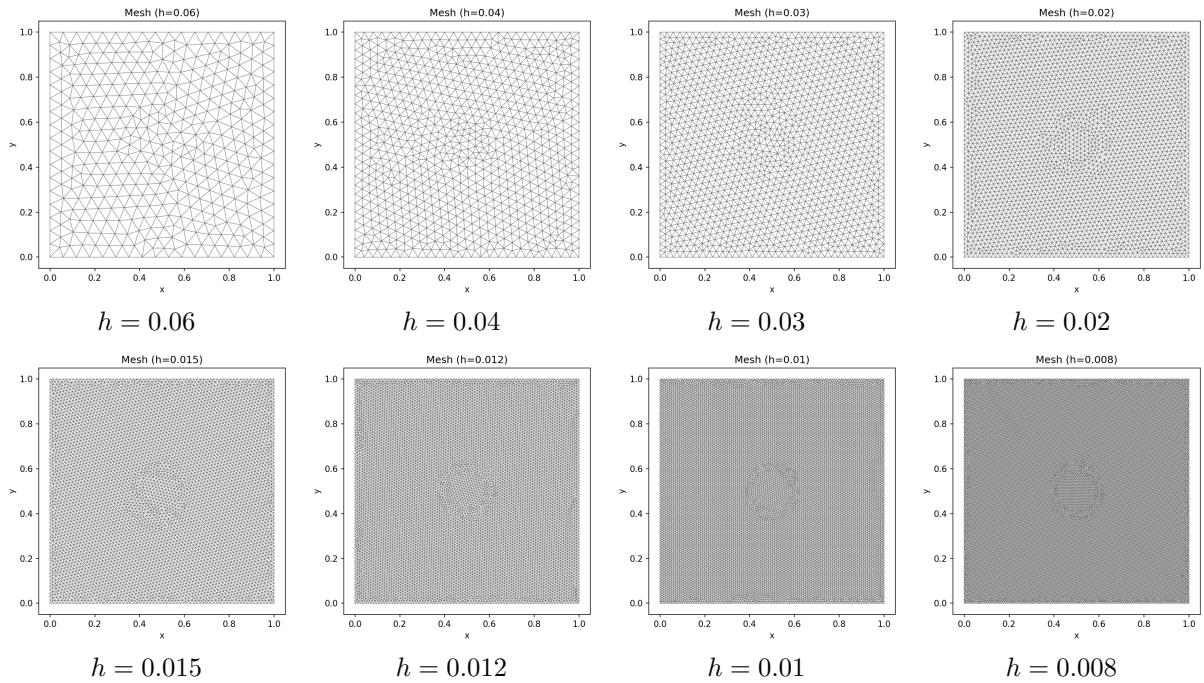

Figure 4: Meshes used in the mesh-independence study.

The resulting objective values and solution norms are summarized in Table 2. Under refinement, the quantities stabilize: for example, the optimal objective decreases from $F(u_*^{(0.06)}) \approx 0.0165171$ to $F(u_*^{(0.008)}) \approx 0.0160695$, while $\|u_*^{(h)}\|_U$ and $\|y(u_*^{(h)})\|_{L^2}$ approach limiting values. Figure 5 shows the same trend graphically, and also confirms that all methods reach essentially the same final objective on each mesh.

h	DOFs	$F(u_*^{(h)})$	$\|u_*^{(h)}\|_U$	$\|y(u_*^{(h)})\|_{L^2}$
0.06	402	0.0165171	2.549350	0.042344
0.04	820	0.0163319	2.607212	0.044949
0.03	1434	0.0162550	2.632510	0.046133
0.02	3093	0.0161709	2.654889	0.047219
0.015	5392	0.0161253	2.665354	0.047746
0.012	8438	0.0161035	2.671025	0.048028
0.01	11899	0.0160847	2.675265	0.048244
0.008	18497	0.0160695	2.678685	0.048419

Table 2: Per-mesh summary of objective and norms. DOFs denotes the number of P1 degrees of freedom.

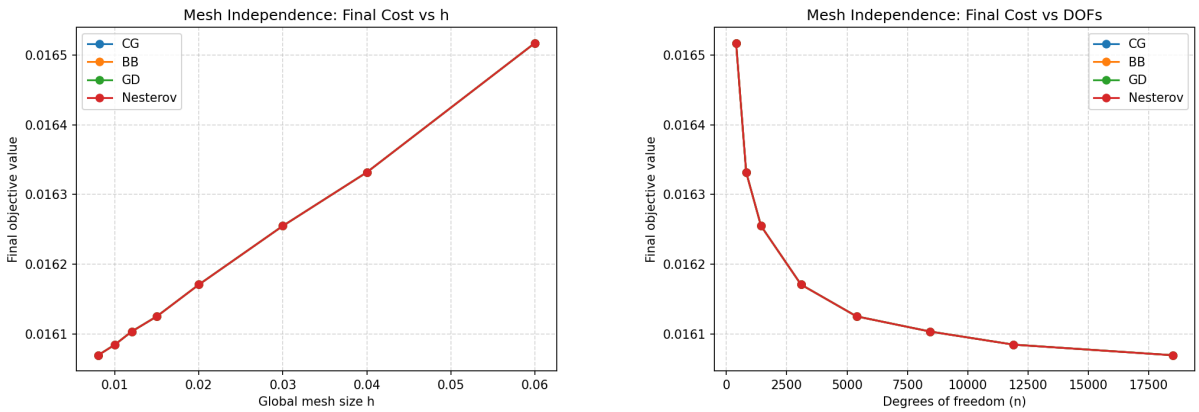

Figure 5: Mesh independence study: final objective value versus mesh size h (left) and versus degrees of freedom (right), for all methods (CG, BB, GD, Nesterov).

Finally, the per-mesh convergence curves (Figure 6) and the iteration summary (Table 3 and Figure 7) indicate that the iteration counts are largely insensitive to the number of degrees of freedom in this experiment: BB terminates after 4 iterations for all meshes, GD after 11–13, Nesterov after 5–6, and CG requires about 18–21 Krylov steps to solve the stationarity system.

h	CG (u_*)	BB	GD($1/L$)	Nesterov
0.06	7	2	6	3
0.04	8	2	6	3
0.03	7	2	6	3
0.02	6	2	6	3
0.015	6	2	6	3
0.012	6	2	6	3
0.01	6	2	6	3
0.008	6	2	6	3

Table 3: Iterations per mesh: CG iterations to obtain u_* by normal equations, and iteration counts for first-order methods.

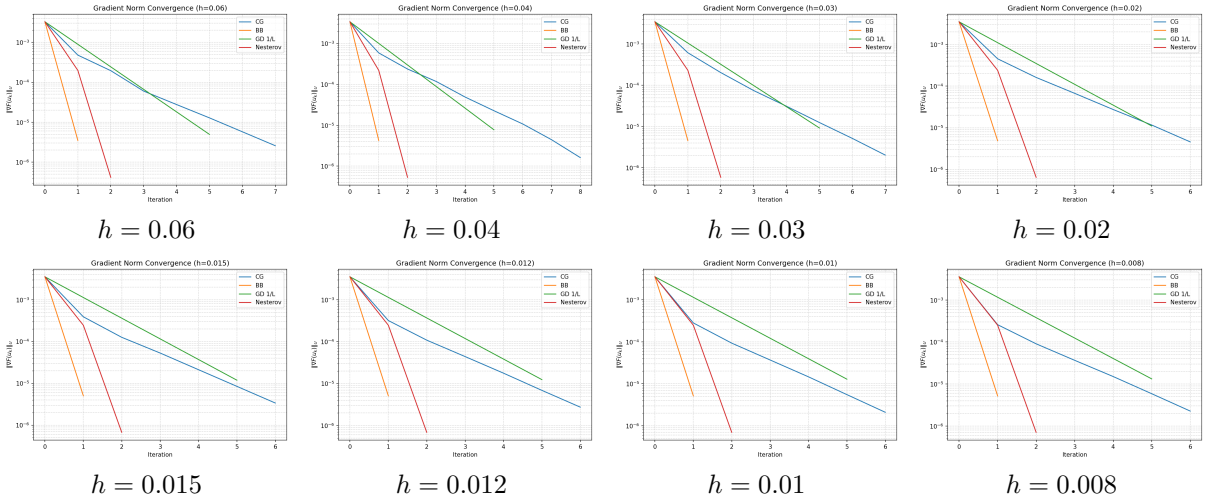

Figure 6: Gradient norm convergence $\|\nabla_U F(\mathbf{u}_k)\|_U$ for each mesh (shown in a 4×2 grid).

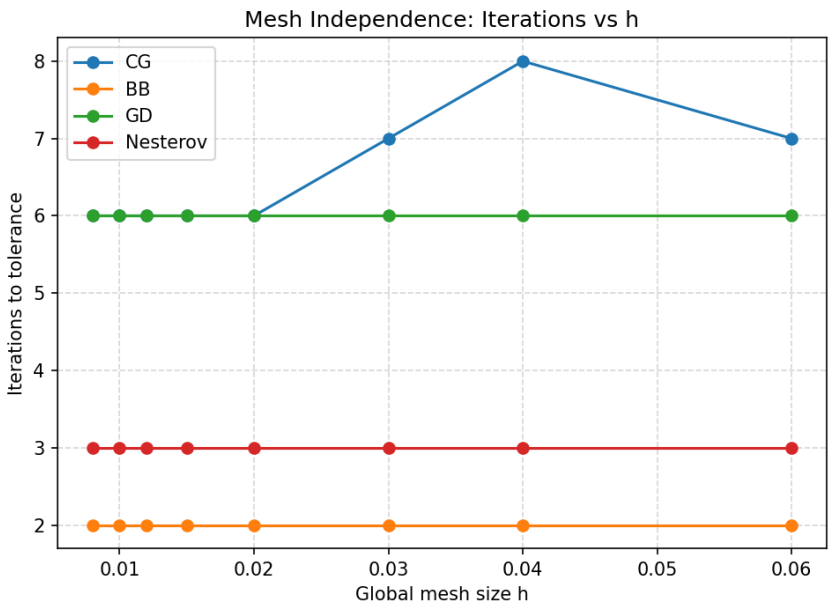

Figure 7: Mesh independence study: iteration counts versus mesh size h (CG iterations for the stationarity solve and first-order iterations for BB/GD/Nesterov).